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Gender diversity on the board of directors

Enhancing firm value or just for show?

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Abstract

This paper investigates the relationship between gender diversity on the board of directors and firm value for S&P500 firms between the years 2012 and 2021. Gender diversity is represented by four different variables which consider gender homogeneity, share of female and the balance between the genders. The paper uses an empirical approach of a 2SLS panel regression with simultaneous equations and fixed effects. The results indicate a significant positive relationship between gender diversity and firm value. This leads to a discussion of the reasoning behind these empirical results and the lack of complete board gender diversity in some of the world's biggest firms. The discussion leads to a recommendation by the author for firms to include more female directors on their boards, but to be wary of the potential difficulty of finding candidates with previous experience and the disadvantages following that.

Gender disclaimer: This paper only considers two genders; male and female, as these are the only genders disclaimed by the chosen data source. Bloomberg only provides the number of female directors and the total number of directors. The share of male directors is in fact the share of non-female directors. If any directors on the boards of the included firms do not identify with either of these two categories, this is unfortunately unknown to the author.

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1. Introduction

In a world where gender equality has become a defining issue during the last decades, the question of whether women should also be in leading positions in the world of business has sparked a heated debate. Some argue for gender quotas on the board of directors to change society's view of how a director should look and be, whereas others believe that it will weaken the economic value to choose directors based primarily on their gender. Studying whether the gender diversity of a board of directors indeed has an effect on firm value is the aim of this paper. Therefore, the following research question is stated:

Does gender diversity on the board of directors have a positive influence on firm value?

The research question poses a hypothesis of a positive relationship between the two. This is based on previous theories and findings in the literature, which will be elaborated through the course of this study.

The paper is organized as follows. Section 2 contains a literature review of previous papers on the subject of gender diversity's effect on firm value. Section 3 takes an interdisciplinary approach to the subject and explains the background for the hypothesis tested in this paper based on existing theories. In section 4, the description of the data sample and its variables are unfolded while the methodology is unraveled in section 5. This leads to the presentation of the results in section 6. Section 7 discusses the reasoning behind the results with an angle on ESG and gender quotas. Finally, section 8 concludes and leaves the reader with a recommendation of how the findings can be applied in real world settings.

2. Literature review

Several studies have dealt with how board composition affect firm value. Carter et al. (2003) investigates diversity as both gender and minorities share of board seats in Fortune 1000 firms. They use Tobin's q as their proxy for firm value and as their diversity measures, they use the share of females or minorities in the boards as well as a dummy variable which takes the value 1, if the board have any minorities or women present. Firstly, they do a matching method, where they match pairs of similar firms where one has a high degree of diversity, while the other have a low degree. They then compare the difference in means of distinctive variables and find that the group with high gender diversity also have higher share of minorities on their board as well as a higher Tobin's q. Secondly, they perform a 2SLS regression with simultaneous equations which further fastens their conclusion that larger diversity leads to higher firm value. Miller & del Carmen Triana (2009) expands on the

findings of Carter et al. (2003) by connecting board demographic diversity to innovation and firm reputation. They then test by an OLS regression if these parameters have a positive effect on firm performance. As performance and not value is in center of their research, they use return on investments and return of sales as their dependent variable. They expand their diversity variable by adding the BLAU index (See section 4.2) as their main variable of interest, to make sure diversity is measured as equal part of each group and not the volume of minority groups present. They find a positive correlation between board diversity and innovation through R&D initiatives, but none with reputation. Reguera-Alvarado et al. (2017) and Campbell & Mínguez-Vera (2008) both use panel data to do a 2SLS regression with simultaneous equations. Both studies use four variables (Percent, dummy, BLAU, and SHANNON index¹) to investigate gender diversity on Spanish boards. Their results follow the line of a positive relationship between diversity and Tobin's q.

Some studies however do not find this positive relationship. Rose (2007) studies the largest firms in Denmark with a cross sectional regression with yearly staggered observations. The study includes only gender diversity and finds no evidence of a positive correlation between diversity and firm value, using the same proxies for both variables as Carter et al. (2003).

This study aims to contribute to the literature with data from recent years, while following the empirical approach of most of the previous literature by using a 2SLS regression with simultaneous equation.

3. Theory

3. Why gender should influence firm value - an interdisciplinary approach

Before the empirical methodology is instigated, the question of why the composition of the board of directors should have an influence on firm value should be addressed. For this an interdisciplinary approach is applied. This section dives into central theories from organization theory, economics, and social psychology to build a theoretical foundation for the hypothesis tested in this paper. None of the theories gives a clear indication of whether the relationship between gender diversity and firm value should be positive or negative, but aids in understanding why there should be a correlation between the two. The four theories used for this purpose are: Resource dependency theory, human capital theory, agency theory and social psychological theory.

¹ See section 4.2

3.1 Resource dependency theory

Resource dependency theory is the study of how external resources of an organization have an effect on the behavior inside said organization. First formalized by Pfeffer & Salancik (1978) who claim that boards connect the corporation to other external organizations in order to address environmental dependencies. Pfeffer & Salancik (1978) propose four primary benefits for the external connections: (1) They provide the firm with outside knowledge and expertise; (2) They work as a channel of communication with external operators of importance to the firm; (3) They link external organizations or operators to the firm and therefore creates a committed support system in the external environment; and (4) they create legitimacy for the firm in the external environment. Hillman et al. (2000) uses these four benefits to create a taxonomy of different type of directors and the influence of their presence on a board. The taxonomy includes insiders, as well as three types of outside directors: business experts; support specialists; and community influential.

They suggest that a more varied board provide further resources to the firm as each type bring different strengths. Resource dependency theory provides one of the strongest theoretical backgrounds to why gender diversity on boards should impact the firms value (Carter et al., 2010).

3.2 Human capital theory

As resource dependency theory fastens that the firm rely on the resources of board members, human capital theory is the foundation of these resources. It claims that each individual comes with a unique set of resources (human capital) such as expertise, skill, information, and potential connections to other external organizations. Occupational differences of the directors reflect the heterogeneity in this capital. This is in line with Hambrick & Mason (1984) that suggest that the characteristics of executives have an influence on the decision and strategies they chose to implement or recommend. The argument for gender diversity is further strengthened in Hillman et al. (2002), where it is shown that female directors most often come with different resources than male. Women directors are more likely to have a higher education than their male counterparts but are less likely to be business experts driven by experience. Further female directors are more prone to follow the advice of and agree with support specialists and community influentials. They also join other boards at a faster rate than their male colleagues, which further reinforce their external connections. This fact could from another perspective add to the discourse that women are only used as token to reinforce legitimacy. This could be a feasible argument as i.e., ESG scores get more and more attention from investors etc. However, the findings of Elstad & Ladegard (2012) offers a different perception and argues that women more often are key influencers rather than tokens. Another discourse which

should be addressed is “the claim that women lack the ‘right’ human capital for directorships” (Terjesen et al., 2009). The human capital evidence of women indicate that this claim has no bearing and women have several performance enhancing qualities i.e., education (Carter et al., 2010). This discourse could however still be the reason why less women are seen in board of directors as they meet several prejudices from the deciding parties (Terjesen et al., 2009).

3.3 Agency theory

One of the most common theories used to explain the relationship between the composition of the board of directors and firm value is agency theory. In an agency framework the board has the important role to control and monitor managers (agents) to resolve agency problems and create value for the shareholders (the principal). A key element of agency theory is the independence of the board. Outside directors should have no incentive to collude with managers, as they wish to build a reputation of being great monitors. The board then only works in the interest of the principals (Carter et al., 2003). The question remains how gender diversity will impact the board seen from an agency theoretical view? One argument should be that a more diverse board will bring different views to the table and therefore be a more activist board. A minority group could be seen as the ultimate outsider, working only to build thier monitor reputation and increase shareholder value. The opposite argument is that minority groups are less likely to be heard and therefore have less influence. This will be elaborated in section 3.4. Carter et al. (2010) claim that high equity ownership by directors would be a stronger incentive than independence to monitor, as they then move closer to the principals. Even though, agency theory has many insides into the effect of the board it has no clear prediction on how gender diversity should affect firm value. It however does not rule out benefits of diversifying.

3.4 Social Psychological theory

The social psychological concept of minority and majority status is derived from social impact theory. This theory argues that in group settings individuals with majority status have larger influence on group decisions (Westphal & Milton, 2000). This advocates the reasoning that the initiatives of diverse directors will have less weight. This is in line with the findings of Elstad & Ladegard (2012) whose results show that woman directors perceive that they receive a larger share of relevant information and have more influence as the gender ratio increases. The effect of diversification is then stronger when women are no longer the minority and there is balance between the two genders. Campbell & Mínguez-Vera (2008) argue that a more gender diverse board leads to varied opinions

and critical thinking which can make decision making inefficient. The theories of group dynamics predict ambiguous effects of gender diversity on firm value.

In summary, as the theoretical framework does not present a definite relationship between gender diversity on the board and firm value an empirical approach is used in this paper to further study the correlation between the two.

4. Data Collection and variable definitions

To impose an empirical methodology a relevant sample, needs to be obtained. Bloomberg is used as the main data source for this paper as it is one of the biggest providers of financial data in the world. To ensure good data quality for my sample I decide to include the largest firms in America as they are more likely to have public information about the structure of their board of directors as well as the other financial data needed. The S&P 500 index is used as it includes 500 large-cap firms from all the leading sectors (*S&P 500®*, 2022). The index further includes some of the world's most well-known firms, which is also considered as their boards are under scrutiny from all over the world. A panel data approach is used to collect the data. This means for each variable needed (see the complete list in table 1a in the appendix) annual data from 2012-2021 is gathered. If one or more variables have no data available for the year, the line is removed from the panel. This is done to ensure high data quality. The final sample is an unbalanced panel with 4782 observations from 497 different firms.

4.1 Dependent variable - Tobin's q as a proxy for firm value

Popularized by James Tobin in the 1970s, Tobin's q ratio has since been seen to be an important measure of firm value. It is defined to be the ratio of the market value of a firm to the replacement cost of its assets. A $q > 1$ indicates the firm creates value for its shareholders as the market value of the firm's assets are larger than its replacement cost (Fu et al., 2016).

In this paper I choose to use the approximate Tobin's q, as estimated by Chung & Pruitt (1994). This decision is due to the efficiency and simplicity of the computation of the approximate Tobin's q, which makes it possible to calculate with only four variables. All which are easily obtained on a firm's balance sheet. Approximate Tobin's q is calculated as seen in equation (1).

$$\text{Approximate } q = (MVE + PS + DEBT)/TA \quad (1)$$

MVE is the firm's market value computed by the product of the firm's share price and number of outstanding common shares, PS is the liquidating value of outstanding preferred stock by the firm in question, DEBT is the value sum of both short- and long-term debt net of the firm's short-term assets. Finally, TA is the book value of the firm's total asset.

The validity of the approximation is shown in Chung & Pruitt (1994) as at least 96.6% of the variability of Tobin's q is explained by this approximation. The approximate q has been used by Al-Slehat, (2019); Carter et al., (2003); Matzler et al., (2005) among others which further endorses its reliability as a good proxy for firm value. Another upside to Tobin's q is that it does not change due to difference in accounting conventions. This means it is valid across different industries. This paper will refer to the approximate q as Tobin's q for simplicity. The accounting information needed to calculate Tobin's q is obtained from the database and added to the sample. Lastly the calculations are done for each firm year and the values for Tobin's q is added to the panel.

4.2 Independent variables - Diversity

As this paper wishes to study whether gender diversity in the board of directors' influence Tobin's q, it is needed to define an independent variable, which captures this said diversity. In this study four different independent variables are used to have a broad perspective on gender diversity.

Firstly, I define a dummy DWOMEN, which takes the value 0 if the board have zero women directors at the given time and 1 if they have one or more. This dummy is created to study whether the change away from gender homogeneity influence the firm value.

Next is the variable PWOMEN which represent the share of women present on the board measured in percent in the given year. PWOMEN gives a clearer view of how diverse the composition of the board really is. Table 1 shows the average share of female directors the last 10 years. Here a clear tendency of an increasing diversity of the board each year visible.

Table 1: Average share of female directors per year

Year	Mean %	N
2012	15.55	450
2013	16.62	460
2014	17.87	466
2015	19.12	474
2016	20.38	477
2017	21.78	482
2018	24.26	488
2019	26.22	493
2020	28.43	496
2021	30.32	496

Further it is investigated whether the share of female directors varies across different industries. Table 2 shows the average share of female directors across industries. The firms are divided into their respective industry using the Bloomberg BICS level 1 tag. To test whether these means differ from the mean of the whole sample a T-test is conducted. 5 out of 11 industries have a p-value below 0,05, which means that it is possible to reject the null hypothesis, that the difference in means is 0. Out of these five the financial sector shows the biggest representation of female directors, with a mean of 25,2%, while Real Estate and Consumer Discretionary have the smallest. However the main conclusion of this investigation is that even with some differences in mean, there are not sectors which are miles apart from the others.

Table 2: Average share of female directors per sector

Sector	Mean %	N	P-value of T-test
Communications	22.29	223	0.797
Consumer Discretionary	20.93	531	0
Consumer Staples	22.22	351	0.921
Energy	21.95	235	0.431
Financials	25.21	596	0
Health Care	21.78	619	0.067
Industrials	21.8	662	0.053
Materials	22.2	235	0.989
Real Estate	20.33	296	0
Technology	21.6	750	0.005
Utilities	23.71	284	0

The PWOMEN variable do however only account for the share of women but not if there are balance between the two genders. Therefore, another variable the BLAU index is used to account for this one-sided diversity of the PWOMEN variable. The BLAU index is calculated: $BLAU = 1 - \sum_{i=1}^n P_i^2$, where n is the number of gender categories present on the board and P_i is the percent of a given gender category. The BLAU index will therefore take the value 0 if only one gender is represented and 0.5 if there are an equal amount of both genders on the board (Miller & del Carmen Triana, 2009).

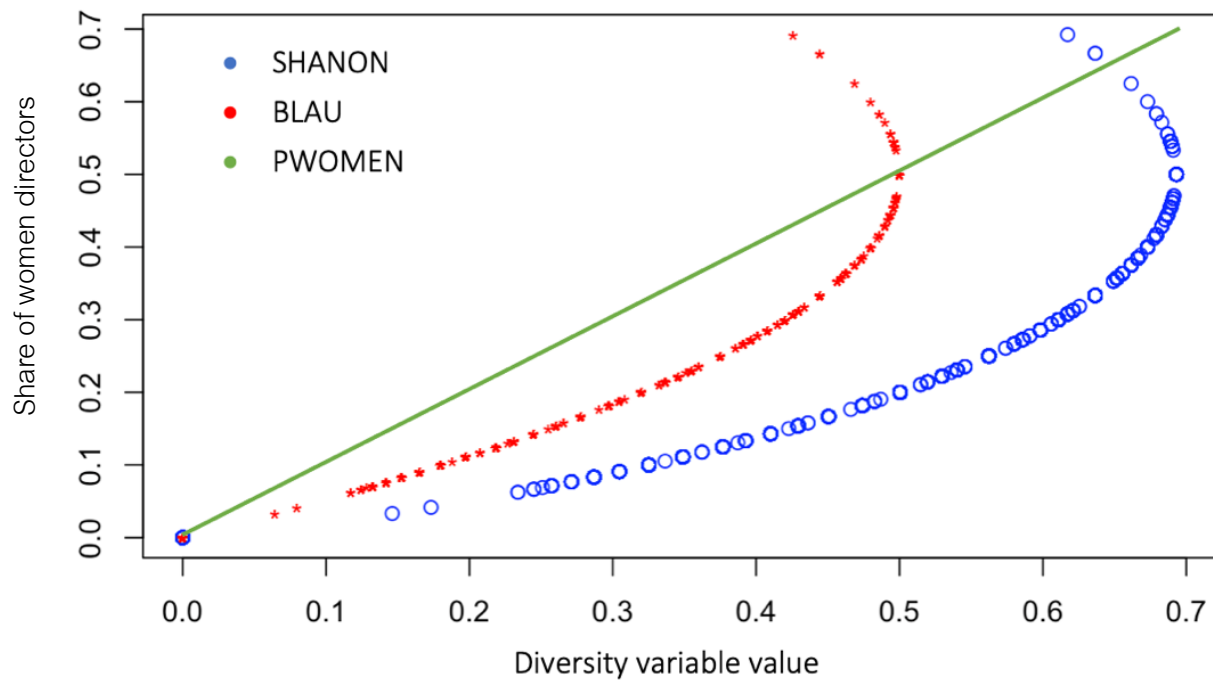
The last measure is the SHANON index, which is similar to the BLAU, but is more sensitive to small changes in composition of the board. This is due to the logarithmic form of the measure, which is calculated $SHANON = -\sum_{i=1}^n P_i \cdot \ln(P_i)$. Here the notation is the same as above. The measure has a less intuitive maximum of 0.69 but remains 0 when only one gender is included. Both these measures are added according to Miller & del Carmen Triana, (2009) and Reguera-Alvarado et al. (2017) and make sure both provide balance and variety into the diversity variables.

Table 3: Independent variables descriptive statistics

	Mean	Standard deviation
DWOMEN	0.96	0.20
PWOMEN	0.22	0.10
BLAU	0.32	0.11
SHANNON	0.49	0.15

Table 3 shows that 96% of all firm years have at least 1 female director on their board. The average share of women is 22%. Across all firm years. The relationship between the three continuous diversity variables is shown in figure 1. This is displayed to give the reader a better understanding of the values the indexes take in different ratios of women directors. The two indexes have the curved form as if the share of women exceeds 50%, the index will take a lower value, as this means less diversity than if the ratio was 50/50. Both indexes therefore take the highest value as the share of women directors are 50%.

Figure 1: Differences in diversity measures



4.3 Control variables

Besides the main variables of interest some control variables are added to the dataset. They are chosen based on previous studies, which indicates they are explaining at least some of the variation in Tobin's q. Board size is added due to the findings of Mak & Kusnadi (2005) whom suggest a

negative relationship between board size and Tobin's q. Additionally Vafeas,1999) finds a negative relationship between the number of annual board meetings and firm value. The natural logarithm values of both variables are therefore added as controls. Further a dummy which takes the value 1 if the directors receive stock compensation, and 0 otherwise, is further added as a control. This is based on the claims of Perry et al. (1999) that stock-based compensation plans for directors provide incentives to monitor management. This promotes the claim that boards have a direct influence on firm value. Peng (2004) demonstrate a positive relationship between outside directors and firm value, which leads to adding percentage of independent directors on the board to the dataset. The average age of the board is added as age is another parameter of diversity, and different measures of diversity seems to be correlated with each other.

Lastly, following Carter et al. (2003) by adding a CEO/chair duality dummy as well as controlling for firm size. Here the return on assets and the natural logarithm of total assets are used as proxies.

4.4 Descriptive statistics

Sample statistics are provided in table 4. The companies are chosen because of their large-cap status, it is however visible from the first line in table 4 that this category is broad measured in terms of total assets. The sample includes all from firm years with 106 million dollars to 3,743 billion dollars in total assets.

Table 4: Sample Statistics

	Min	1st Quart.	Median	Mean	3rd Quart.	Max
Total Assets	106.00	7,078.00	18,242.00	68,892.39	45,980.00	3,743,567.00
Tobin's q	0.05	1.00	2.00	2.21	3.00	23.00
Board Size	5.00	9.00	11.00	10.86	12.00	30.00
Women on Board	0.00	2.00	2.00	2.44	3.00	9.00
Average Age	43.86	61.00	63.00	62.77	65.00	78.00
Return on Assets	-61.82	2.00	6.00	6.71	10.00	76.00
Annual Meetings	1.00	6.00	7.00	8.18	10.00	43.00
Independent Directors	3.00	8.00	9.00	9.17	11.00	26.00

With the difference in mean and median it can be deduced that the top firms are in a league of their own. The sample have an average Tobin's q of 2.21. Further the 1. quartile of the sample has a q above one, which indicates that the market value of the firm is greater than the book value of their

assets. The firms have a median board size of 11, and these boards have an average of 2.44 female and 9.17 independent directors. These directors attend an average of 8.18 annual meetings. The minimum and maximum value of average age has quite a large spread of just above 29 years. It is clear that both these values are extremes as from the 1. quartile at 61 years to the 3. quartile at 65 years there are only a 4-year difference. From this information it is not possible to say anything on an individual level of the directors, but table 4 shows that most of the firm years have a board with an average age somewhere in their sixties. Some firm years shows a negative return on assets with a minimum of -61.82. The mean return of assets is 6.74.

5. Methodology

This section aims to elaborate the implementation of the chosen methodology to provide insight into the relationship between gender diversity in the board of directors and firm value.

The fundamental model tested is a fixed effects panel 2SLS regression done with a within transformation. The within transformation is chosen to ensure more degrees of freedom. This is done in line with the studies of Campbell & Mínguez-Vera (2008). Time fixed effects is added to take into account the trends which are specific for the different years. This could be market tendencies for stocks or the larger focus on diversity developed over the last 10 years, which is very clear in table 2. Individual firm effects are added to account for different characteristic of the firms. This could be the distinct difference in total asset from the largest to the smallest firm. By adding the individual fixed effects these outliers will have less of an effect. The first model is represented by equation 2:

$$Tobin's\ q = \alpha + \beta_1 Diversity_{i,t} + \sum \beta x_{i,t} + \delta_t + \mu_i + \varepsilon_{i,t} \quad (2)$$

Where *Diversity* is one of the four previously mentioned diversity variables and *x* are a vector consisting of the following control variables: MEETINGS; BOARDSIZE; CEO; STOCK; INDEPENDENT and TA as described in section 4.3. δ_t and μ_i are time and firm fixed effects respectively. Lastly, ε is the error term. This results in four different specifications of the model.

To consider the possible endogeneity between the diversity variables and Tobin's q proposed by Westphal & Milton (2000) the second model is made and solved as a system of simultaneous equation. The second model is described by equation 2:

$$Diversity = \alpha + \beta_1 Tobin's\ q + \sum \beta z_{i,t} + \delta_t + \mu_i + \varepsilon_{i,t} \quad (3)$$

Here the independent and dependent variables are swapped. z includes the following controls: CEO; STOCK; INDEPENDENT; AGE and ROA.

To confirm the efficiency of the fixed effects model compared to the random effects model, a Hausman test is performed on each of the four specifications of the two models. Further a Breusch-Pagan test and a Honda test performed to assess the presence of unobserved heterogeneous effects. To report on heteroscedasticity the Breusch-Pagan test is used and to inspect for serial correlation the Breusch-Godfrey and Durbin-Watson is applied. Newey-West standard errors are then used to account for any serial correlation or heteroskedasticity present in the error term.

6. Empirical results

In this section the results of the 2SLS regression are presented. In table 5 the results of the model with the variable PWOMEN as the independent variable is showed. The regression provides a positive coefficient on a 0,1% significance level. This indicates that a higher share of women directors has a positive influence on firm value as described by Tobin's q . Further it is seen that the natural logarithm of annual meetings all has significant negative relationship with firm value. This is expected according to Vafeas (1999). However, the negative significant results for independent directors were not expected, as it contradicts findings of both Peng (2004) and Carter et al. (2003). Both the CEO duality and the stock compensation dummy shows no significant results. If the attention is shifted to the second model in table 5, a significant positive relationship between independent directors and women directors is visible. This could be due to the fact that women more often are support specialist or community influentials rather than business experts (Hillman et al., 2002).

It can also be observed that there is a significant negative relationship between high average age and share of women. This indicates that a younger board is more likely to choose a more gender diverse set of directors. Further Tobin's q has no significance on the share of women directors, which is in the line with the findings of Campbell & Mínguez-Vera (2008).

Table 6, which presents the results of the dummy variable DWOMEN show the same positive effect of diversity in the form of at least one female director on the board and firm value. It further shows a significant negative relationship between Tobin's q and board size, which is in line with previous literature. This relationship is however only significant in this version of the regression.

Table 5: PWOMEN

	Tobin's q - PWOMEN	
	(1)	(2)
PWOMEN	20.000*** (3.500)	
MEETINGS	-0.560*** (0.097)	
BOARDSIZE	0.360* (0.190)	
Tobin's q		0.003 (0.005)
AGE		-0.350*** (0.043)
CEO	-0.078 (0.072)	0.0003 (0.003)
STOCK	-0.220 (0.280)	0.019** (0.007)
INDEPENDENT	-4.300*** (0.900)	0.230*** (0.020)
TA	-0.760*** (0.120)	
ROA		0.0002 (0.001)

Notes: *** Significant at the 1 percent level.
 ** Significant at the 5 percent level.
 * Significant at the 10 percent level.

Table 6: DWOMEN

	Tobin's q - DWOMEN	
	(1)	(2)
DWOMEN	21.000*** (2.900)	
MEETINGS	-0.530*** (0.086)	
BOARDSIZE	-3.000*** (0.510)	
Tobin's q		-0.120*** (0.015)
AGE		-0.370*** (0.140)
CEO	0.100 (0.068)	-0.021** (0.010)
STOCK	-0.980*** (0.280)	-0.052 (0.034)
INDEPENDENT	-5.800*** (0.920)	0.310*** (0.087)
TA	-0.880*** (0.120)	
ROA		0.016*** (0.002)

Notes: *** Significant at the 1 percent level.
 ** Significant at the 5 percent level.
 * Significant at the 10 percent level.

In table 7 and table 8 the results of the models using the diversity index BLAU and SHANNON as the independent variable is displayed. The results of the two are as expected very similar. They both show a positive relationship between gender diversity and firm value. The coefficient for the SHANNON index is smaller than that of the BLAU. As Shannon penalize small negative changes in diversity harder, this could indicate that the exact composition of the genders is less important than that diversity is present at all.

Table 7: BLAU			Table 8: SHANNON		
	Tobin's q - BLAU			Tobin's q - SHANNON	
	(1)	(2)		(1)	(2)
BLAU	19.000*** (3.200)		SHANNON	16.000*** (2.600)	
MEETINGS	-0.470*** (0.084)		MEETINGS	-0.470*** (0.083)	
BOARDSIZE	0.027 (0.200)		BOARDSIZE	-0.260 (0.220)	
Tobin's q		-0.010* (0.005)	Tobin's q		-0.025*** (0.007)
AGE		-0.350*** (0.049)	AGE		-0.440*** (0.070)
CEO	-0.094 (0.072)	-0.002 (0.004)	CEO	-0.081 (0.071)	-0.004 (0.005)
STOCK	-0.390 (0.280)	0.009 (0.009)	STOCK	-0.480* (0.280)	0.005 (0.014)
INDEPENDENT	-4.900*** (0.940)	0.250*** (0.024)	INDEPENDENT	-5.200*** (0.970)	0.330*** (0.036)
TA	-0.830*** (0.120)		TA	-0.860*** (0.120)	
ROA		0.002*** (0.001)	ROA		0.004*** (0.001)
Notes: *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.			Notes: *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.		

Table 6-8 all have a small but significant negative coefficient of Tobin's q on the respective diversity variable, which differs from the results of the PWOMEN regression as well as findings of other literature. The rest of the three tables show similar relationships as displayed as in table 5.

The Hausmann tests return a very small p-value for all four specifications of the models, which confirms that the fixed effects model is the most efficient. The results of these can be seen in tables 2a-5a in the appendix as well as the results of all tests for serial correlation, unobserved heterogeneous effects, and heteroscedasticity. The key takeaway from these tests is the need to apply the Newey-West standard errors, which is done in all four specifications of the 2SLS regressions.

7. Discussion

The results presented in section 6 shows a clear positive relationship between gender diversity on the board of directors and firm value. This is consistent with previous findings of (Campbell & Mínguez-Vera, 2008; Carter et al., 2003; Miller & del Carmen Triana, 2009). Even though the literature advocates a positive empirical relationship the reasoning could be found in a correlation rather than a causal relationship. Birindelli et al. (2018) shows that there is a positive correlation between the share of women directors and ESG ratings in the banking sector. The reason for this can be that women directors have a larger interest in the ESG area, but also that companies which is focused on improving their ESG score is more likely to employ women on their boards, as this in itself will lead to a higher rating. According to the findings of Ginglinger & Raskopf (2019) the first is most often the case. They use evidence from the implementation of gender quotas in France to investigate the development in firms' ESG focus. Their findings suggest that women directors have human capital in form of experience, qualities and preferences which leads to the implementation of more ESG-oriented policies. The question remains why this should influence firm value? As sustainability becomes a hotter topic in the world of investors and consumers the firm value, especially when expressed by Tobin's q should increase in firms with a better ESG rating. This is consistent with the findings of Fatemi et al. (2018) and Yu et al. (2018) which both show that better ESG disclosure and ratings have a positive influence on firm value. The ESG argument points to women directors influencing the firm value positively through sustainable initiatives. Table 2 shows a clear tendency to employ a larger share of women directors each year over the last decade, but the average ratio is still less than one third women directors. How come this is still the case if the evidence encourages a gender balanced board? One reason could be the history of women being less active in the workforce and especially in leading positions. Following that the potential bias women meet by employers about their human capital as discussed in section 3.2 further weakens their chances to get the spot on the board. To break this spiral some countries have introduced gender quotas on board of directors. These quotas are a controversial topic among many as an argument against could be that if the candidate shall be a woman, it could possibly lead to the best candidate being turned away because of his gender. As less women have experience sitting on a board it can be harder for firms to find experienced female candidates for the seats. The findings of Ahern & Dittmar (2012) supports this theory. They look at mandated gender quotas in Norway and find that they have a negative effect on firm value and attribute it to among other factors the increase in younger and more inexperience directors. Greene et al. (2020) looks at the effect of Californian mandated gender quotas from

2018 and find that companies which have a lower supply of female candidates is hit harder on firm value. These findings in some way contradicts the empirical evidence of this paper as gender quotas increase diversity which should increase firm value. One reason why there can be validity in both these two findings can be found by incorporating the theories presented in section 3. As less women have board seats it is harder to employ experienced female directors, and they then have a harder time fulfill the third and fourth “benefits of directors”² as expressed by Pfeffer & Salancik (1978). The lack of plenty experienced women also explains Hillman et al. (2002) findings that women directors join multiple boards quicker than their male counterparts as they are somewhat of a rare species. If diversity on boards is only positive when the minority groups bring in prior experience this leads to a “chicken or egg” dilemma. If companies do not let women into their boards, they will never get the experience, but then the companies cannot reap the benefits of a diverse board of directors. The positive findings of diversity on firm value in this paper could further be due to the choice of sample. When only including S&P500 firms it would not be unreasonable to assume that they have access a larger pool of experienced female candidates, than smaller and lesser-known companies. If the same positive relationship is visible in a sample of smaller firms is out of the scope of this paper but would be an interesting subject for further studies.

8. Conclusion and recommendation

Gender diversity and equality have become more prominent on western societies’ agendas over the last decades. But the question whether it contributes only with ethical prosperity or if it indeed could increase the economic wealth is still not without an ambiguous answer.

This paper investigates whether gender diversity on the board of directors in S&P500 firms leads to higher firm value. Through an empirical approach using a fixed effects 2SLS regression with simultaneous equations this paper finds a significant positive relationship between different diversity variables and Tobin’s q. These findings are substantiated with theories from different fields to attempt explaining the reasoning behind the empirical results. This paper further aims to provide a perspective for shareholders to take into account when they have to vote for or nominate new candidates for board seats. The recommendation of the author is to try diversifying the board of directors as much as possible to reap the benefits and attempt to prevent one gender to be the minority group as this leads to better communication and a broader range of viewpoints. It should however be considered

² (3) They link external organizations or operators to the firm and therefore creates a committed support system in the external environment; and (4) they create legitimacy for the firm in the external environment.

that these benefits could be dependent on not only the gender of the possible candidates, but also the level of experience. The positive relationship between gender diversity and firm value is not certain to exist if the candidates are not the most qualified for the position. Shareholders which are concerned about ESG issues should have additional incentive to vote for or nominate female candidates, even with less experience, as they are said to bring ESG-oriented policies on the agenda. Further a gender diverse board is perceived as good governance in an ESG taxonomy and therefore it will bring a higher rating to the company.

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Appendix

Table 1a: Variable definitions

Dependent variable	
Tobin's q	Proxy for firm value
- MVE	Market value computed by the product of the firm's share price and number of outstanding common shares
- PS	Liquidating value of outstanding preferred stock
- DEBT	The value sum of both short- and long-term debt net of the firm's short-term assets
Independent variables	
DWOMEN	Dummy variable for at least one woman present on the board
PWOMEN	Ratio of women on board in percent
BLAU	Diversity measurement
SHANNON	Diversity measurement
Control variables	
TA	The natural logarithm of total assets
ROA	Return on assets
CEO	Dummy for CEO duality
MEETINGS	The natural logarithm of number of annual board meetings
STOCK	Dummy variable takes 1 if a stock-based compensation plans for directors is in place
INDEPENDENT	Ratio of independent members on the board in present
BOARDSIZE	The natural logarithm of the size of board described by number of directors
AGE	The natural logarithm of the average age of board members

Table 2a: Tests of PWOMEN regression

Test	Statistic	P-value	Conclusion
Fixed vs. Random Effect			
Hausman	426.236	0e+00	Fixed effect
Individual Effect			
Honda	73.232	0e+00	Individual effect
Serial Correlation			
Breusch-Godfrey	1762.1279	2.803802e-296	Serial correlation
Durbin Watson	0.9792	1.404026e-273	Serial correlation
Heteroskedasticity			
Breusch-Pagan	200.4226	9.340471e-40	Heteroskedasticity

Note:

This table shows the test performed on the PWOMEN regression. The results of the tests are presented by their statistics, P-values and interpretation.

Table 3a: Tests of DWOMEN regression

Test	Statistic	P-value	Conclusion
Fixed vs. Random Effect			
Hausman	349.5926	0e+00	Fixed effect
Individual Effect			
Honda	78.994	0e+00	Individual effect
Serial Correlation			
Breusch-Godfrey	1664.3062	2.312534e-276	Serial correlation
Durbin Watson	1.0277	1.975644e-248	Serial correlation
Heteroskedasticity			
Breusch-Pagan	178.6592	3.738986e-35	Heteroskedasticity

Note:

This table shows the test performed on the DWOMEN regression. The results of the tests are presented by their statistics, P-values and interpretation.

Table 4a: Tests of BLAU regression

Test	Statistic	P-value	Conclusion
Fixed vs. Random Effect			
Hausman	396.017	0e+00	Fixed effect
Individual Effect			
Honda	72.5471	0e+00	Individual effect
Serial Correlation			
Breusch-Godfrey	1739.4391	1.185242e-291	Serial correlation
Durbin Watson	0.9899	6.753121e-268	Serial correlation
Heteroskedasticity			
Breusch-Pagan	190.8435	9.949541e-38	Heteroskedasticity

Note:

This table shows the test performed on the BLAU regression. The results of the tests are presented by their statistics, P-values and interpretation.

Table 5a: Tests of SHANNON regression

Test	Statistic	P-value	Conclusion
Fixed vs. Random Effect			
Hausman	356.2329	0e+00	Fixed effect
Individual Effect			
Honda	72.2044	0e+00	Individual effect
Serial Correlation			
Breusch-Godfrey	1724.0521	1.617526e-288	Serial correlation
Durbin Watson	0.9974	5.582072e-264	Serial correlation
Heteroskedasticity			
Breusch-Pagan	200.4226	9.340471e-40	Heteroskedasticity

Note:

This table shows the test performed on the SHANNON regression. The results of the tests are presented by their statistics, P-values and interpretation.